Wildfire Risk Durably Falls After Timber Harvests:
Evidence from 40 Years of Harvest and Fire in Canada

Jeffrey E. Sun*

Preliminary draft. All errors are my own.

Link to Latest Version

July 20, 2026

Abstract

I estimate the association between timber harvest and subsequent wildfire risk nationally across Canada's managed forest using a staggered-adoption event study on 447935 thirty-metre stands which cover the universe of Canada's managed forest. Smaller studies have shown mixed results, often that harvested stands burn more, but they are typically confined to a single region and to plot- or severity-scale ecological evidence. To my knowledge, this is the first national, continental-scale quasi-experimental estimate of the harvest–fire relationship in Canada. Harvested stands face a first-fire hazard about 28% lower than comparable unharvested neighbors. The protective effect is persistent to at least 29 years after harvest. My preferred specification controls by strictly pre-treatment siting and terrain covariates, ecozone-by-year and fireshed fixed effects. The result is robust to clustering scheme, to dropping the extreme 2023 season, and to choice of covariates up to dropping all covariates: the raw association is already protective and attenuates under adjustment. The result highlights the potential benefits of co-management of timber harvest and wildfire risk, and suggests that the current separate-management regime in Canada may be imposing very large economic, environmental, and human costs.

1 Introduction

In the record 2023 Canadian fire season, wildfires in Canada released approximately 2.4 billion tonnes of CO_2 into the atmosphere: roughly 6% of global fossil-fuel emissions, and more than four times the emissions from all economic activity in Canada combined in that year (Byrne et al., 2024). While these are gross biogenic emissions, partially reabsorbed by regrowth over subsequent years, and the 2023 season does not represent recurring annual emissions, the scale of the externality is nevertheless enormous.

The land that burns is, to a large extent, the same land that Canada manages for timber. Yet in every province, forest and fire management are administered by operationally independent

*Department of Economics, University of Toronto

and often largely non-coordinating agencies. Wildfire suppression is conducted by provincial and territorial agencies, while timber production is planned decades ahead through licensed tenures, often implemented by non-governmental forest management companies (Tymstra et al., 2020).

British Columbia has moved furthest toward closing this gap: its Forest Practices Board has called for a shift from the separate-management status quo toward landscape fire management, in which wildfire objectives are set across land-use zones and fuel treatment is coordinated with forest management (British Columbia Forest Practices Board, 2023).

If timber harvest raises or lowers a stand's subsequent fire risk, then separate management may impose large economic, environmental, and human costs. Scientifically, the question remains open. A widely cited empirical reading holds that harvested land burns more.

I estimate the causal effect of past timber harvest on a stand's subsequent *first-fire hazard*, estimated nationally across Canada's managed forest. This presents two major challenges, which my design addresses to a greater extent than previous studies, though completely solving these may be impossible. First, harvest location is endogenous, and depends on access, stand age, and operability, all of which may be correlated with fire risk. Second, burned areas are endogenous to fire suppression efforts, which are not only targeted, but targeted partly at merchantable timber.

I estimate the association between past harvest and subsequent first-fire hazard with a staggered-adoption discrete-time (complementary log-log) hazard event study fit to the national managed-forest panel. I use separate datasets for treatment and outcome, which I find crucial to avoid a mechanical correlation between the two due to the shared classifier used to detect both harvest and fire in previous studies.My outcome data is an independent record of fire perimeters;the covariates are pre-treatment forest structure and terrain, measured before the harvest year; the fixed effects are additive ecozone-by-year and coarsened-fireshed absorbers; the controls are never- and not-yet-treated stands; and inference is clustered on fire-years against a $t(G-1)$ reference. The ecozone-by-year fixed effects control for the common fire weather of a given season in a given biome (e.g. 2023 in the boreal). The fireshed fixed effects are spatial tiles on the order of a hundred kilometers and focus the analysis to comparisons among neighbors that share a fire environment rather than across the continent. The estimating panel covers 447935 stands and about 12.18 million stand-years,[1] spanning 39 fire-years. I defer the design to Section 3.

Conditioning on siting through observables and fixed effects, harvested stands show a robust protective association with subsequent first-fire hazard. The pooled estimate is a cloglog coefficient of -0.3332—a hazard ratio of 0.7166 (95% CI [0.565, 0.908], $p \approx 0.007$), a first-fire hazard roughly 28% below that of comparable unharvested stands. The association is protective at 27 of 29 post-treatment horizons and survives dropping the extreme 2023 fire year. The raw, unadjusted association is *already* protective with a hazard ratio near 0.55 before any adjustment. Conditioning attenuates it toward 0.7166. Currently, however, pre-trend and placebo-lead analysis is challenging due to immortal-time survivorship compounded by salvage-window contamination (Section 5), and

[1] About 8.90 million stand-years enter the preferred specification once fully separated zero-fire cells are absorbed, see Section 3.

I hope to resolve these issues in future versions.

This paper makes three contributions. First, it provides the first national, continental-scale quasi-experimental estimate of the harvest–fire relationship in Canada: a staggered-adoption first-fire-hazard event study fit to the universe of the managed forest, rather than the plot-scale or single-region evidence that dominates the existing literature. Second, it measures the fire outcome independently of the harvest treatment—an NBAC fire-perimeter record produced by a pipeline separate from the CanLaD disturbance classifier—and clusters inference on fire-years, so that the estimate rests on measurement independence and on inference appropriate to spatially and serially correlated fire rather than on the pixel-level standard errors of a shared-classifier design. Third, it connects the reduced-form estimate to the economics of fire–forest co-management (Section 6), the policy margin that motivates the question. The quantified welfare and management model that co-management would require is future work, which I do not claim here.

1.1 Related Literature

There is a large forest-ecology literature on the effect of harvest on fire risk, and this literature review will necessarily omit many important and foundational contributions, providing instead only a small sample. Lindenmayer et al. (2009) find that logging elevates fire proneness in moist forests. A long line of stand- and severity-scale studies debates whether the young, even-aged regeneration that follows a cut carries fire more readily than the standing biomass it replaced. My contribution against this body is one of design and scale: a national, quasi-experimental hazard estimate built on an independent fire outcome, rather than plot-scale or severity-conditional ecological evidence, and one that estimates a sign for clearcut-dominated harvest as practiced rather than for an idealized treatment.

A related strand studies deliberate fuel management, where the mechanism my event-time profile is meant to adjudicate is stated cleanly. Agee and Skinner (2005) set out the principles of fuel-reduction treatment, in which removing ladder and surface fuel lowers fire behavior; Thompson et al. (2020), by contrast, find in an experimental boreal crown fire that recent crown thinning reduced neither the rate of spread nor total fuel consumption, a caution that even a deliberate treatment need not lower the hazard. The tension—young post-harvest fuel structure raising flammability against reduced standing load lowering it—is precisely the one a clearcut, which is not a designed fuel treatment, leaves unresolved a priori. That the sign is theoretically ambiguous is what makes it worth estimating. Against this literature I contribute the empirical sign, and the note that a clearcut and a fuel treatment are not the same intervention.

The result must be identified against a rising and increasingly correlated fire baseline. Hanes et al. (2019) document fire-regime intensification in Canada over the last half century, and Ellis et al. (2022) the climate-driven decline in fuel moisture behind it; together they are why harvest is at most a marginal lever on a climatically driven trend, and why I cluster inference on fire-years and probe the estimate by dropping the 2023 mega-shock. On the policy side, Tymstra et al. (2020) review the Canadian wildfire-management status quo and the co-management opportunity my mechanism discussion builds toward. Finally, the estimator sits within the modern heterogeneity-robust difference-in-differences literature—Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Borusyak et al. (2024) for estimation under staggered adoption; Goodman-Bacon (2021) and de Chaisemartin and D'Haultfœuille (2020) for the negative-weighting critique of naive two-way fixed effects; and Roth et al. (2023) for the pre-trend and robustness practice I follow. My contribution here is not a new estimator but a disciplined application of these tools at continental scale, with inference appropriate to spatially and serially correlated fire.

2 Data

2.1 Sources

Treatment (timber harvests) comes from the Canadian Landsat Disturbance dataset (CanLaD), a set of annual 30 m disturbance rasters covering 1985–2024 that classify each pixel-year into one of a small set of disturbance types—wildfire, harvest, windthrow, water, pest, and defoliation (Perbet et al., 2022). Harvest is my treatment. Its combination of annual resolution and national coverage enables a high-resolution, powered staggered-adoption event study feasible. One limitation is that CanLaD is itself a product of Landsat change-detection rather than a ground census. In particular, it labels the disturbance type but not the intent behind it, so that harvest of salvage timber after a fire is coded identically to routine commercial harvests. In Section 5, however, I find that this likely does not matter for the main estimates. I also use CanLaD's wildfire code to study how the use of a contaminated fire measure may bias the analysis, and I plan to use the windthrow code for a natural-disturbance comparison in a later falsification exercise.

The outcome is measured by a different instrument: the National Burned Area Composite (NBAC), which assembles fire perimeters from agency fire mapping and satellite imagery through a pipeline entirely separate from the CanLaD disturbance classifier (Canadian Forest Service, 2024). This is important: the fire outcome and harvest treatment, if sourced from the same classifier, would be linked mechanically, introducing correlated measurement error.

I transform the NBAC perimeters into the hazard model's outcome: a stand's first fire. For each stand and each post-treatment year I record whether an NBAC perimeter first intersects it that year; a stand that has not yet burned remains in the risk set, and once it burns it leaves. The event study is thus a discrete-time model of the time to first fire, and the object of interest is how harvest shifts that hazard. Importantly, reburns—second and later fires on the same ground—are not modeled. The set of already-burned stands is already small, and the set of stands with second burns is smaller still; they would also represent a plausibly qualitatively different forest state into the model and muddy the identification. The survival panel is right-censored: a stand leaves the risk set when the observation window ends without a fire. The estimator accounts for both exits.

Pre-treatment covariates describe the land as it was before the cut. Each is chosen because it plausibly predicts where operators choose to harvest. This is the siting margin that the design must confront. Forest structure variables are aboveground biomass, canopy height, and canopy closure from SCANFI, a spatialized rendering of the Canadian National Forest Inventory built on dense Landsat time series (Guindon et al., 2024). These three variables proxy for the level of development and current fuel load of a stand; I expect them to be highly correlated with merchantable value. Terrain variables are elevation, slope, and a terrain-ruggedness index from the Canadian Digital Elevation Model; it serves as a proxy for (1) ease of access for harvest operations and (2) the physical determinants of fire spread. A prior-fire indicator records whether a stand had already burned within the observation window before treatment, in which case a stand is held out of the main specification.Each covariate is measured before harvest, as structure or terrain variables affected by the harvest might otherwise serve as a mediator of the effect of harvest on fire (Section 3).

Non-forest, water, and off-Canada cells are masked out using the VLCE2 land-cover product. The disturbance and fire records then cover the entirety of Canada's managed forest. The study population post-filtering is 447935 stands—221830 treated and 226105 control—distributed across 201 firesheds spanning the national managed forest. Firesheds are tiles on the order of 100 km; they serve as the level of geographic fixed effects which force comparisons to be made among "neighbors" rather than across the continent (Section 3).

A significant endogeneity issue is that harvests are spatially concentrated on more operable, less remote, and more merchantable land, which may burn less readily than the remote, rugged, or unmanaged forest that is never cut. My pre-treatment covariates and fixed effect design exist to address exactly this potential confounding effect. Indeed, the raw harvest–fire association is already protective before any adjustment, and conditioning attenuates this effect (Section 4), suggesting that the raw effect is partially spurious, but that controls attenuate this siting bias.

The magnitudes size the problem the rest of the design confronts. Table 1 sets treated and control stands side by side on the six pre-treatment covariates, and the gap is concentrated in forest structure. Harvested stands average 126 tonnes per hectare of aboveground biomass against 74 for controls, stand 17.0 meters tall against 11.0, and close 62% of canopy against 46%—standardized differences of 0.71, 0.92, and 0.89, between seven-tenths and nine-tenths of a pooled standard deviation. Terrain is far closer to balanced: the standardized differences are -0.14 on slope, $+0.06$ on elevation, and -0.15 on the terrain-ruggedness index. The picture is the siting story of the previous section: the land that gets cut is the more developed, more merchantable forest, and it differs from the never-cut remainder along precisely the structural axes that also govern how readily a stand burns. This imbalance is what the covariate adjustment and fixed effects of Section 3 are built to remove, and its size is a first indication of how much work that adjustment must do.

Table 1: Descriptive statistics for treated (harvested) and control (never- and not-yet-harvested) stands. The upper block reports, for each pre-treatment covariate—SCANFI forest structure (aboveground biomass, canopy height, canopy closure) and DEM terrain (slope, elevation, terrain-ruggedness index), all measured strictly before the harvest year on the common 30 m reference grid (Section 2.2)—the pooled, treated, and control mean with its standard deviation in parentheses, and the standardized mean difference (treated minus control over the pooled standard deviation). The lower block reports the panel scale: stands, first-fire events, and total and estimation-sample stand-years, each split by treatment arm. The standardized differences quantify the siting imbalance that the covariate adjustment and fixed effects of Section 3 are built to remove.

	All	Treated	Control	Std. diff.
Aboveground biomass (t/ha)	99.5 (78.1)	125.7 (81.7)	73.7 (64.8)	+0.71
Canopy height (m)	13.9 (7.2)	17.0 (6.2)	11.0 (6.8)	+0.92
Canopy closure (%)	54.2 (19.8)	62.2 (13.8)	46.2 (21.6)	+0.89
Slope (degrees)	5.1 (6.9)	4.6 (5.7)	5.6 (7.9)	−0.14
Elevation (m)	536.5 (378.1)	548.7 (365.7)	524.5 (389.5)	+0.06
Terrain ruggedness index	7.1 (10.0)	6.4 (7.9)	7.9 (11.6)	−0.15
Stands	447,935	221,830	226,105	
First-fire events	55,655	12,848	42,807	
Stand-years	12,402,337	4,302,342	8,099,995	
Stand-years used	12,180,507	4,080,512	8,099,995	

2.2 Grid Assembly and Spatial Harmonization

The design requires that, for each 30 m × 30 m forest cell, I observe its harvest history, its subsequent wildfire exposure, and its pre-treatment characteristics *at the same physical location*—but these come from separate national raster datasets (CanLaD annual disturbances for treatment, NBAC fire perimeters for the independent outcome, SCANFI for forest structure, the Canadian Digital Elevation Model for terrain, VLCE2 for land cover), each distributed on its own map projection, grid origin, and resolution. A given point on the ground lands at different pixel coordinates in each dataset, so the layers cannot simply be stacked—the spatial analogue of merging panels whose identifiers are coded on incompatible keys. "Warping" fixes this: using GDAL's `gdalwarp`, I reproject and resample every source layer onto one common reference grid, the CanLaD 30 m grid spanning forested Canada (178 400 columns by 119 100 rows), interpolating continuous variables such as elevation bilinearly and categorical ones such as land cover by nearest neighbor, with no-data and off-Canada cells coded consistently so that a location absent from one layer is absent from all. Pixel (i, j) then denotes the identical patch of ground in every layer, and only then are the datasets mergeable into the cell-level panel the event study consumes. It is a one-time but heavy operation—the national grid holds roughly 21 billion pixels ($178{,}400 \times 119{,}100$)—which is why it dominates the pipeline's setup cost.

2.3 Unit of Analysis

The unit of analysis is a 30 m sampling cell drawn from a strided lattice, not the full pixel grid. A panel over all 178,400 × 119,100 pixels would carry on the order of 10^{10} observations, and, worse than being unwieldy, would treat adjacent 30 m pixels cut in a single operation as near-duplicate observations rather than as independent information. I therefore subsample the national grid onto a spatial lattice. Within each read band I keep every hundredth pixel column, which places sampled columns about 3 km apart —a stride of 100 pixels at 30 m resolution—and I read only a periodic subset of the latitude bands, so the lattice thins in both directions while remaining an unbiased spatial sample of the managed forest. The wide ∼3 km column spacing is the feature that matters for the stability assumption: two cells cut in one harvest operation are essentially never both sampled along a row, so the lattice does not stack near-identical neighboring pixels as though they were independent draws. Rows within a sampled band are read at full 30 m resolution, so some residual spatial dependence remains; that is handled at the inference stage, where standard errors are clustered on fire-years rather than on cells (Section 3). Cells enter the hazard likelihood unweighted, so the estimand is the average treatment effect on the average retained cell rather than on the average hectare or the average harvest operation. For readability I refer to these sampling cells as “stands” through the rest of the exposition, and the scale macros—447935 and the like—count them.

Two timing conventions close the construction. CanLaD dates a harvest to the year it is first detected, which for a late-season cut can trail the operation on the ground by a year; to keep that ambiguity out of the estimates I quarantine event time $\tau = 1$ and trust the panel only from $\tau \geq 2$ onward. I then follow each stand out to $\tau = 30$ years after treatment. Beyond the near horizons the treated sample thins as stands are progressively censored by their own first fires and by the edge of the observation window, so the longest-horizon estimates rest on comparatively few stands and are read with that attrition in mind (Section 4). The resulting panel—stands in event time, an independent NBAC fire outcome, and strictly pre-treatment covariates on a common grid—is the object that the event study of Section 3 consumes.

3 Design and Identification

3.1 Estimand and Estimating Equation

The quantity I estimate is the average treatment effect on the treated, resolved as a profile in event time. For a stand harvested in calendar year c, write $\tau = t - c$ for the years elapsed since the cut; the object of interest, ATT(τ), is the effect of having been harvested on the stand’s probability of a first NBAC fire τ years later, relative to the fire experience of otherwise-comparable stands not yet cut. Because the outcome is a first fire and a stand leaves the risk set once it burns, the natural summary is a hazard rather than a level: I model the discrete-time hazard h_{it}, the probability that stand i records its first fire in year t given that it has survived fire-free to the start of that year. The profile $\{\mathrm{ATT}(\tau)\}$ traces how harvest shifts that hazard across the post-treatment horizon; a single pooled post-treatment contrast collapses it to one number, and it is the pooled number, properly powered, that I treat as the headline and the profile that I read for shape.

I fit the profile with a staggered-adoption discrete-time complementary-log-log hazard,

$$\mathrm{cloglog}\{h_{it}\} \;=\; \sum_{\tau\geq 2} \beta_\tau \, D^\tau_{it} \;+\; x_i'\gamma_1 \;+\; (x_i \odot x_i)'\gamma_2 \;+\; \alpha_{e(i),\,t} \;+\; \phi_{f(i)}, \tag{1}$$

where D^τ_{it} is an indicator that stand i stands τ years past its own harvest in year t, x_i is the vector of strictly pre-treatment SCANFI structure and DEM terrain covariates described in Section 2, entered as linear and quadratic terms (x_i and its elementwise square $x_i \odot x_i$, with coefficient vectors γ_1 and γ_2); $\alpha_{e,t}$ is an additive ecozone-by-year fixed effect, and ϕ_f is a coarsened-fireshed fixed effect. The complementary-log-log link is the discrete-time counterpart of a proportional-hazards model: $\exp(\beta_\tau)$ is a hazard ratio, the multiplicative shift in a harvested stand's first-fire hazard at horizon τ relative to the reference, so that a coefficient below zero (a hazard ratio below one) is a lower subsequent fire hazard.[2] The event-time arms run from $\tau = 2$ out to $\tau = 30$. I quarantine $\tau = 1$ because CanLaD dates a harvest to the year it is first detected and a late-season cut can trail the operation on the ground by a year (Section 2.3); reading the panel only from $\tau \geq 2$ keeps that one-year labeling ambiguity out of the estimates. For the pooled headline I replace the arm-by-arm indicators with the single post-treatment indicator $\mathbf{1}\{\tau \geq 2\}$, so that one coefficient carries the average protective shift over the whole post-treatment window and inherits the full power of the panel.

Cells enter the likelihood unweighted, so the estimand is the average treatment effect on the average retained cell rather than on the average hectare. The comparison group is built to be clean in the sense the staggered-adoption literature now insists on: the controls are never-treated and not-yet-treated stands only, never already-harvested land. A not-yet-treated stand contributes to the contrast at horizons before its own eventual cut and exits the control pool at the point it is treated, so no stand is ever compared against its own post-treatment self, and the negative-weighting pathologies of naive two-way fixed effects under staggered timing (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020) do not arise.

The two fixed effects do complementary work, and neither is decoration. The ecozone-by-year terms sweep out the common fire weather of a given season in a given biome—the boreal in 2023, say—so that the harvest contrast is never drawn between a stand in a burning year and a stand in a quiet one. The fireshed terms, spatial tiles on the order of 100 km that group nearby stands sharing a fire environment, force the comparison to be made among neighbors rather than across the continent: within a fireshed, harvested and unharvested stands face the same terrain, the same regional climate, and the same suppression jurisdiction, and it is the within-fireshed difference in fire experience that identifies β. What that spatial absorption costs the estimand—which stands

[2] I estimate Equation (1) in R with `fixest::feglm` under the `cloglog` link, absorbing the two high-dimensional fixed effects by alternating projection; the sufficient-statistic aggregation and the separation-induced fixed-effect drop are detailed in Section B.

actually contribute once the fixed effects are in—is the subject of Section 3.4.

3.2 Identifying Assumptions and Threats

The identifying assumption is conditional parallel trends: within a fireshed and a given ecozone-year, and after adjusting for pre-treatment forest structure and terrain, the first-fire hazard of harvested stands would have moved in parallel with that of comparable unharvested stands in the absence of harvest. Under that assumption the post-treatment divergence in Equation (1) recovers the ATT profile. The assumption is not directly testable at the horizons that matter; its natural check is the pre-treatment lead profile, and I report that check—including the ways it is compromised as currently built—in Section 5.5. Two threats bear on whether the assumption holds, and observable adjustment meets them unequally. I take them in turn.

The first and most consequential is siting. Harvest is not assigned but chosen: operators cut where access, stand age, and operability make it worthwhile, on flatter and more reachable ground carrying merchantable timber, and that same land tends to be land that would burn less readily than the remote, rugged, or unmanaged forest that is never cut. A raw harvest–fire comparison therefore contrasts systematically different parcels, and it does so in a known direction: treated land is intrinsically the safer land. The pre-treatment covariates and the fireshed fixed effect exist to absorb exactly this imbalance—the covariates by conditioning on the structure and terrain that drive both the decision to cut and the propensity to burn, the fireshed terms by holding the comparison to neighbors—but I do not claim they absorb it fully. Their partial success is visible in the estimates: the raw association is already protective, and conditioning moves it *toward* the null, a pattern I read as confounding being removed rather than an effect being manufactured, and take up in full in Section 5.4. Figure 1 shows the same thing in the covariates: the raw standardized differences on forest structure run from 0.71 to 0.92, and demeaning within the ecozone-year and fireshed fixed effects pulls them down to near 0.28 without closing them to zero. That the within-fixed-effect gap does not close to zero is the visible sign that the adjustment is partial: a same-signed siting confound not captured by structure, terrain, or fireshed would leave a residual protective bias, and I return to bounding it in Section 5.5 and Section 5.6.

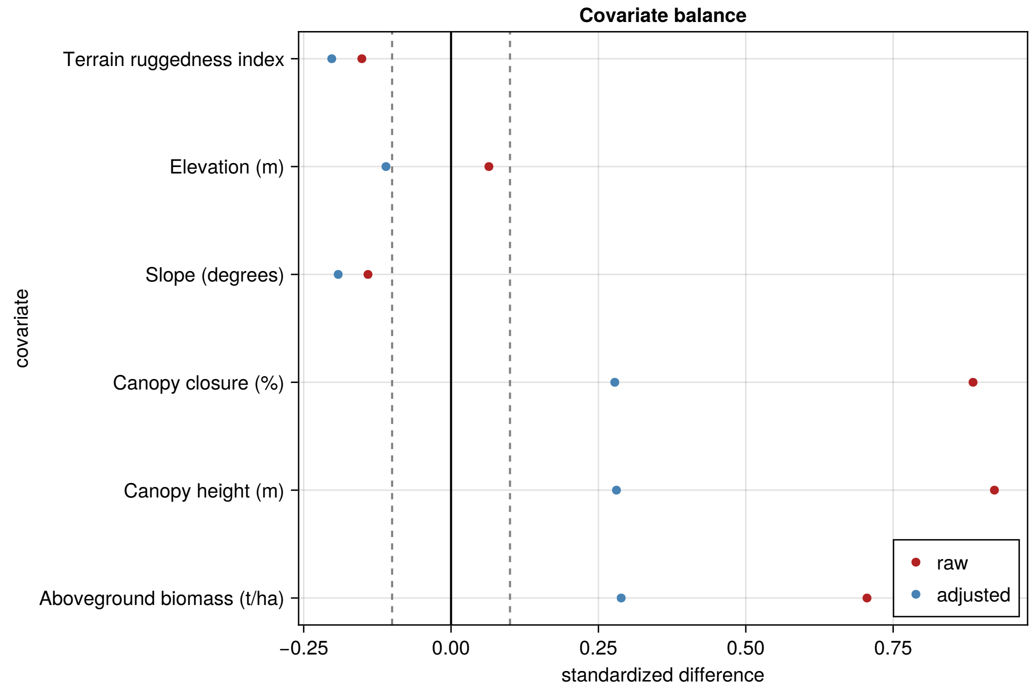

Figure 1: Standardized differences in pre-treatment covariates between treated (harvested) and control stands. Each row is a covariate—SCANFI forest structure (aboveground biomass, canopy height, canopy closure) and DEM terrain (slope, elevation, terrain-ruggedness index). The raw markers give the standardized mean difference (treated minus control over the pooled standard deviation) in the unadjusted stand-level sample; the adjusted markers give the within-fixed-effect difference after demeaning by ecozone×year and coarsened fireshed on the estimation sample. Dashed lines mark ±0.1. Raw standardized differences on forest structure run from 0.71 to 0.92; the within-fixed-effect differences are near 0.28. Terrain differences are small throughout.

The second threat is suppression. Fire suppression is itself targeted, and targeted partly at merchantable timber, so a stand that is harvested may also be a stand that is defended, and the protection I read off the hazard could in part reflect the fire agency's choices rather than the cut's physical legacy. The direction this pushes the estimate is not obvious—suppression concentrated near timber would bias the ATT down, toward over-protection, while the road and ignition access that harvest brings would push it the other way—and I do not resolve it here. A management-zone split is the first-class response, and its reach is limited by the absence of a national management-zone raster; I treat that gap plainly in Section 5 and Section 7 rather than assume it away.

Beyond these two threats, a point about measurement is worth making. Reading the fire outcome off the same disturbance classifier that defines the harvest treatment would correlate the two before any physical process on the ground intervenes—a mechanical artifact of construction rather than a fact about fire. Drawing the outcome from NBAC, a fire record produced by an entirely separate pipeline (Section 2.1), avoids that by construction. I treat measurement independence as sound practice, not as a claim about what generated any particular earlier finding.

It is worth being explicit about what the design settles and what it does not. Measurement independence and the inference regime are in hand: the fire outcome is drawn independently of the harvest treatment, and inference is clustered on fire-years against a $t(G-1)$ reference that survives fireshed and two-way schemes (Section 3.3, Section 5.1). Siting-selection is not. The pre-trend machinery that would test conditional parallel trends at the horizons that matter is non-functional as currently built—an immortal-time survivorship gradient in the never-treated-only leads reference, compounded by salvage-window contamination (Section 5.5)—so conditional parallel trends is assumed here, not verified, and a same-signed siting confound that observable adjustment cannot reach, potentially a material share of the estimated protection, cannot be excluded. This is why I hold the claim to a robust protective *direction* rather than an identified *effect*, and set out the agenda that would settle it as a forward program in Section 5.5. I state the boundary once, here, as scope.

3.3 Inference

Fire is spatially and serially correlated on a scale that dwarfs the stand. A single season's weather lights up whole biomes at once, and 2023 lit up the boreal from one side of the country to the other, so the relevant independent draws are not stands—of which there are hundreds of thousands—but fire-years, of which there are 39. I therefore cluster the standard errors on fire-years and refer the resulting t statistic to a $t(G-1)$ distribution with $G = 39$, the conservative small-cluster reference that guards against the over-rejection a normal approximation would invite when the number of clusters is modest. Treating the correlated pixels of a fire season as independent draws is precisely the error that makes conventional standard errors in a regional, pixel-level design far too small to believe (Section 1); clustering on fire-years is the first-order correction, and it is the primary specification throughout.

Firesheds, the 100 km spatial tiles intersected with ecozone that serve as the spatial fixed effect, also furnish an alternative clustering block, and I report fireshed-clustered and two-way (fire-year $\times$ fireshed) standard errors as robustness in Section 5.1. The point estimate is identical across all three schemes by construction; only the interval width moves, and all three intervals exclude a hazard ratio of one. I do not read the tighter fireshed p-value as the headline: fire-year clustering is the conservative, defensible primary, because the season-wide shock is the correlation structure most likely to be underestimated, and I let the more cautious number lead.

3.4 The Estimand's Support

The fireshed and ecozone-year fixed effects buy their spatial control at a price the estimand must pay openly. A fixed-effect group that contains no fire at all is perfectly separated in a hazard model—its cells contribute no information to a first-fire likelihood and are dropped—and in a country where most stand-years never burn, that is a large share of the panel. The two absorbers together drop 26.9% of stand-years, with 127 of 390 ecozone-year groups (32.6%) and 86 of 296 fireshed groups (29.1%) fully separated by the absence of any fire within them; among treated rows specifically, 29.2% drop for the same reason. The estimand is therefore the ATT *among fire-experiencing cells*—stands in ecozone-years and firesheds that burn at all—not the ATT over the entire managed forest. I state this as scope rather than defect: it is the price of forcing the harvest comparison to be drawn among neighbors that share a live fire environment, and it is the population over which a within-fireshed hazard contrast is even defined.

Within that population, I ask how much of the treated stock actually sits on common within-fireshed support, and the answer is nearly all of it. Of the 201 firesheds in the overlap diagnostic, 197 contribute to identification (98.0%), holding 99.5% of treated stands on within-fireshed support; the four non-contributing firesheds are tiny treated-only or zero-on-support cells, each with at most seven treated stands, immaterial to the estimand. On that basis I report the headline as a national ATT rather than a local sliver. The qualification I attach, and do not drop, is that this is a *marginal* common-support test: it certifies that treated stands' covariate *ranges* lie inside the control ranges of their own fireshed—a bounding-box check—and it cannot certify joint support or detect holes in the interior of the covariate space. I describe the result as marginal within-fireshed support, and I do not claim that positivity is a non-issue; the full overlap diagnostic is in the appendix (Section D), and the fixed-effect-drop accounting in Table 6.[3]

4 Results

Conditioning on siting through observables and fixed effects, harvested stands in Canada's managed forest show a robust protective *association* with subsequent first-fire hazard. The pooled post-treatment estimate is a complementary-log-log coefficient of -0.3332, a hazard ratio of 0.7166 (95% CI [0.565, 0.908]), a first-fire hazard roughly 28% below that of comparable unharvested stands, and the association is protective at 27 of 29 post-treatment horizons. The estimand is the ATT among fire-experiencing cells—the fixed effects drop the zero-fire cells (Section 3.4)—and within those, 98.0% of firesheds and 99.5% of treated stands lie on marginal within-fireshed support, so I report it as a national estimate rather than a local one. It is a protective direction, not yet a fully identified effect; the identification limits are the business of Section 5, and every downstream reading inherits them. I present the estimates in that spirit: first the primary event study, then the progressive specification build that shows how much of the result is the raw data and how much is the adjustment, holding the causal reading for Section 5.

4.1 Primary Event-Study Estimates

Table 2 presents the primary estimates, the event-time profile above and the pooled post-treatment arm below the rule. The pooled arm is the powered statement, and it is protective: a hazard ratio of 0.7166 (cloglog -0.3332, SE 0.1171, $t = -2.84$, $p = 0.0071$ against $G = 39$ fire-years), with a 95%

[3] The overlap diagnostic is computed over 201 firesheds, while the fireshed-clustered inference of Section 5.1 uses $G = 296$ fireshed groups; the two counts come from different coarsenings of the fireshed layer and I report each with its own number rather than silently reconciling them.

confidence interval of [0.565, 0.908] that sits entirely below one. Read as a magnitude, harvested stands face a first-fire hazard about 28% lower than otherwise-comparable unharvested neighbors in the same fireshed and ecozone-year. An alternative aggregation that averages the per-horizon hazard ratios rather than fitting a single pooled arm returns a slightly deeper 0.6818; I report both and treat the pooled scalar as the properly-powered headline, since the mean-of-horizons weights the thin, noisy long-horizon arms equally with the well-populated early ones.

The profile in the upper rows of Table 2, plotted as Figure 2, is protective almost everywhere. Across the 29 horizons from $\tau = 2$ to $\tau = 30$, the hazard ratio lies below one at 27, the two exceptions being $\tau = 9$ (HR 1.18) and $\tau = 29$ (HR 1.14), both with confidence bands that comfortably cover one, and $\tau = 13$ essentially at the line (HR 0.97). The near horizons are protective but individually imprecise: $\tau = 2$ through $\tau = 8$ all sit below one, between roughly 0.66 and 0.83, but each carries a band wide enough to cover it, so the early profile is a gentle protective drift rather than a sharp on-impact drop. There is, in particular, no early positive bump—no horizon at which harvested stands burn detectably *more*—which is the feature the mechanism discussion will lean on (Section 6). Precision arrives later and repeatedly: ten individual horizons—$\tau = 11$ (HR 0.55), 17 (0.58), 19 (0.45), 20 (0.61), 21 (0.63), 22 (0.61), 24 (0.48), 25 (0.40), 26 (0.52), and 28 (0.33)—have bands that exclude one on their own, and every one of them is protective. Not a single horizon is individually significant in the wrong direction.

The protection, if anything, tends to *deepen* at longer horizons rather than fade: the most protective single horizon is $\tau = 28$, at a hazard ratio of 0.33, and the run from the late teens through the twenties sits well below one. That deepening is read with its power in view, and not as a strengthening mechanism (Section 6), because the treated sample thins steadily as horizons lengthen: stands are progressively censored by their own first fires and by the edge of the observation window, so the at-risk count falls with the horizon and the confidence bands widen accordingly. The far-horizon points are bounds on a thinning sample rather than sharp corrections. I shade the horizons beyond $\tau \approx 20$ as low-power in Figure 2 and read them with that attrition in mind. The pooled arm, which pools information across the whole window, is the statement that carries weight; the profile tells me the pooled number is not the average of a sign that flips horizon to horizon, but of a protection present throughout that simply sharpens as it accumulates.

Table 2: Primary event-study estimates of the association between past harvest and subsequent first-fire hazard. The upper rows report the event-time profile ATT(τ) for $\tau = 2, \ldots, 30$ (the $\tau = 1$ arm is quarantined for harvest-year timing ambiguity); the final row, set off by a rule and in bold, reports the pooled post-treatment arm ($\mathbf{1}\{\tau \geq 2\}$). Each row reports the hazard ratio $\exp(\beta)$ from the complementary-log-log first-fire hazard and the 95% confidence interval in brackets. The model adjusts for pre-treatment SCANFI structure and DEM terrain and includes ecozone×year and coarsened-fireshed fixed effects, fit by `fixest::feglm`. Confidence intervals are cluster-robust on fire-years with a $t(G-1)$ reference ($G = 39$). The estimand is the ATT among fire-experiencing cells (Section 3.4).

τ	HR $= e^{\beta}$	95% CI (HR)
2	0.81	[0.57, 1.13]
3	0.76	[0.56, 1.02]
4	0.66	[0.35, 1.22]
5	0.67	[0.43, 1.05]
6	0.73	[0.40, 1.34]
7	0.75	[0.42, 1.34]
8	0.83	[0.58, 1.19]
9	1.18	[0.66, 2.11]
10	0.82	[0.41, 1.66]
11	0.55*	[0.32, 0.94]
12	0.82	[0.57, 1.17]
13	0.97	[0.59, 1.60]
14	0.79	[0.57, 1.08]
15	0.78	[0.49, 1.25]
16	0.85	[0.49, 1.46]
17	0.58*	[0.35, 0.98]
18	0.66	[0.38, 1.15]
19	0.45*	[0.23, 0.91]
20	0.61*	[0.38, 0.99]
21	0.63*	[0.40, 0.98]
22	0.61*	[0.44, 0.85]
23	0.59	[0.29, 1.20]
24	0.48*	[0.31, 0.74]
25	0.40*	[0.24, 0.68]
26	0.52*	[0.32, 0.85]
27	0.63	[0.30, 1.29]
28	0.33*	[0.21, 0.52]
29	1.14	[0.55, 2.36]
30	0.95	[0.64, 1.42]
Pooled	**0.72**	[0.57, 0.91]

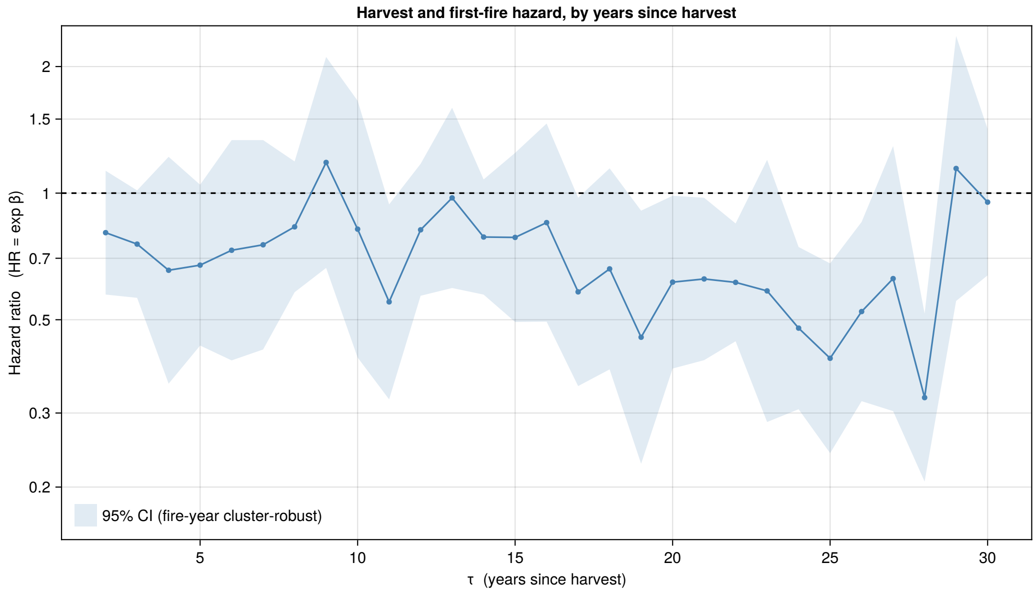

Figure 2: Event-study profile of the harvest–wildfire association. Each point is the hazard ratio $\exp(\beta_\tau)$ for a first NBAC fire at τ years since harvest, relative to never- and not-yet-harvested control stands, from the complementary-log-log first-fire hazard with pre-treatment covariates and ecozone×year and coarsened-fireshed fixed effects. The shaded band is the 95% fire-year cluster-robust confidence interval; the dashed line marks HR $= 1$ (no association). The vertical axis is on a log scale. Points below the line indicate a lower subsequent fire hazard for harvested stands; 27 of 29 horizons lie below one. Confidence bands widen at long horizons as the number of still-at-risk treated stands declines.

4.2 Progressive Specification Build

The specification menu in Table 3 adds controls cumulatively, and it is presented as disclosure of the estimate's conditionality, not as a search across specifications for the one that reports best. Row (C0) is the raw, unadjusted contrast: a hazard ratio of 0.551. Row (C1) adds the Tier-A siting confounders—SCANFI biomass, canopy height, canopy closure, and DEM slope—and the estimate moves to 0.658. Row (C2) adds the terrain precision terms, elevation and the terrain-ruggedness index, and the estimate holds at 0.658—the terrain terms add essentially nothing once the SCANFI structure covariates are in, which locates the confounding work in stand structure and canopy rather than in elevation and ruggedness. Row (C3) adds the ecozone×year and coarsened-fireshed fixed effects and moves the estimate to 0.7166; row (C4), the preferred specification shown in bold, is that same fixed-effects model and carries the 0.7166 headline. The estimate moves *up*, toward one, at every step that adds a control—adjustment attenuates the raw protection rather than deepening it. The count of first-fire events (both arms) in the estimation panel is common across the rows at 54,694, so the movement from (C0) to (C4) is the controls entering and not the sample shifting under them; the rows explain the same fires with progressively more of the siting structure held fixed.

Two features of the row walk I flag here and read in Section 5.4. The first is that movement itself: the estimate travels from 0.55 to 0.72 as siting confounders and fixed effects enter, a direction that carries an identification meaning I take up as the paper's coefficient-path centerpiece. The second is a disclosure about statistical significance. Only the two fixed-effects rows, (C3) and (C4), have confidence intervals that exclude a hazard ratio of one; the raw and covariate-only rows (C0) through (C2) have wider intervals that cover one, at p-values of 0.067, 0.109, and 0.113 respectively. I report that plainly, because it matters for how the preferred row is read: the fixed effects are what sharpen the estimate to conventional significance, and a reader is entitled to know that the raw protective point, while already present, is not on its own distinguishable from one. What the fixed effects add is not the sign, which is there from the start, but the precision—and the interpretation of that pattern is the business of the next section.

Table 3: Progressive-specification build of the pooled harvest–wildfire association. Rows add controls cumulatively: (C0) raw / unadjusted; (C1) + Tier-A confounders (SCANFI biomass, height, canopy closure, DEM slope); (C2) + terrain precision terms (elevation, terrain-ruggedness index); (C3) + ecozone×year and coarsened-fireshed fixed effects; (C4) the preferred specification (shown in bold). Each row reports the pooled post-arm hazard ratio $\exp(\beta)$, its 95% confidence interval, the fire-year cluster-robust standard error, the t statistic, the two-sided p-value ($t(G-1)$, $G = 39$), the number of first-fire events (both arms), and the number of fire-year clusters G. The estimate attenuates toward one as siting confounders and fixed effects enter, the pattern expected when harvest is sited on lower-fire-risk land. The menu is presented as disclosure of the estimate's conditionality, not as specification selection.

Specification	HR	95% CI (HR)	SE	t	p	Events
raw / unadjusted	0.551	[0.29, 1.04]	0.316	-1.89	0.067	54694
+ Tier-A confounders	0.658	[0.39, 1.10]	0.255	-1.64	0.109	54694
+ terrain (safe-precision)	0.658	[0.39, 1.11]	0.258	-1.62	0.113	54694
+ FE (ez_year, fireshed)	0.717*	[0.57, 0.91]	0.117	-2.84	0.007	54694
PREFERRED (continuous, fire-year cluster)	**0.717***	**[0.57, 0.91]**	**0.117**	**-2.84**	**0.007**	**54694**

5 Robustness and Falsification

This section is the paper's center of gravity, and I place it in front rather than behind. It asks, in turn, whether the protective association survives the inference regime, the extreme fire-years, the support restriction, and the progressive adjustment; whether the pre-treatment leads permit a parallel-trends test; how large an unobserved confound would have to be to overturn the result; and where across the country the association concentrates. The answers are mostly reassuring and, in one place, deliberately not: the pre-trend machinery does not license the test I would want, and I say so and bound what follows from it.

5.1 Inference and Clustering

The pooled point estimate does not move with the choice of clustering; only the interval does. Table 4 reports the preferred specification under three schemes. Under fire-year clustering, the primary, the hazard ratio is 0.7166 with $G = 39$ and $p = 0.0071$. Under fireshed clustering it is the same 0.7166 with $G = 296$ and a tighter $p = 0.0046$; under two-way fire-year $\times$ fireshed clustering it is again 0.7166 with $p = 0.0127$. All three intervals exclude a hazard ratio of one. I lead with the fire-year number and do not promote the tighter fireshed p-value, because the season-wide shock—one weather system lighting up a whole biome—is the correlation structure most likely to be underestimated, and fire-year is the conservative block that respects it. The two-way scheme, which allows correlation along both the temporal and the spatial margin at once, is the most demanding of the three, and the estimate clears it as well.

Table 4: Sensitivity of the pooled harvest–wildfire estimate to the choice of clustering. Rows report the same preferred complementary-log-log specification with cluster-robust standard errors computed on fire-years ($G = 39$), on coarsened firesheds ($G = 296$), and two-way on fire-year$\times$fireshed. Columns report the hazard ratio, its 95% confidence interval, the standard error, the two-sided p-value, and the number of clusters G. The point estimate (HR 0.717) is identical across schemes by construction; only the standard error and interval width change, and all three intervals exclude HR = 1. Fire-year clustering is the primary specification (the conservative block for season-wide fire shocks); the others are reported as disclosure.

Clustering	HR	95% CI (HR)	SE	p	G
fireyear	0.717	[0.57, 0.91]	0.117	0.007	39
fireshed	0.717	[0.57, 0.90]	0.117	0.005	296
fireyear $\times$ fireshed	0.717	[0.55, 0.93]	0.127	0.013	39

5.2 Leave-Out Robustness

The record is dominated by a handful of extreme seasons, and 2023 above all: one cross-fireshed mega-shock that burned on the order of seven times a typical year's area (Appendix E.2). A natural worry is that the protective association is merely that season's shadow—that harvested stands happened to sit off the path of the 2023 fires and the estimate is reading geography, not harvest. The leave-2023 test in Table 5 answers it directly. Dropping 2023 entirely and re-estimating on the remaining 38 fire-years leaves the association not weaker but stronger, at a hazard ratio of 0.6307 (cloglog -0.4609, $p = 0.0057$). The sign is not a 2023 artifact; if anything, the single most extreme season was pulling the pooled estimate toward one, not away from it.

A wider leave-out, dropping the entire 2018–2024 high-fire window, moves the estimate further still, to a hazard ratio of 0.554, and I report it in the appendix (Section C, Table 7) and read it asymmetrically. Dropping that window removes about a fifth of the fire-years, cutting G from 39 to 32 and with it the power to detect anything; a *null* in that thinned sample would have been indeterminate rather than a failure, so the strengthened point is consistent with the association but is not independent confirmation of it. I resist reading the monotone ordering—full sample weaker than leave-2023 weaker than leave-2018–24 in protective magnitude—as a mechanism. It is in part a story about which fire-years remain in the sample and how their composition shifts the pooled contrast, not a dose-response in harvest.

Table 5: Leave-out robustness of the pooled harvest–wildfire estimate. Rows report the preferred specification on the full sample (the anchor) and after dropping 2023, the single extreme cross-fireshed fire season (R1). Columns report the hazard ratio, its 95% fire-year cluster-robust confidence interval, and the t statistic. The protective sign persists and strengthens when 2023 is removed, so the estimate is not a 2023 artifact. Further leave-outs (R2, dropping the 2018–2024 high-fire window; R15, +prior_fire) are reported in the appendix (Section C, Table 7).

Specification	HR	95% CI (HR)	t
Full sample (preferred)	0.717*	[0.57, 0.91]	-2.84
leave-2023-out	0.631*	[0.46, 0.87]	-2.93

5.3 Within-Fireshed Common Support

The estimand rests on within-fireshed comparisons (Section 3.4), so the question is whether enough of the treated stock has unharvested neighbors to compare against, or whether the national label hides an estimate identified off a handful of firesheds. It does not. Marginal within-fireshed common support is near-universal: 197 of the 201 firesheds in the overlap diagnostic contribute (98.0%), holding 99.5% of treated stands on within-fireshed support, and the four non-contributing firesheds are tiny treated-only or zero-on-support cells with at most seven treated stands each. The fireshed fixed effect thus identifies the ATT off essentially the whole fire-experiencing treated population, and I report the headline as a national estimate on that basis.

The limit I keep attached to that statement is what the diagnostic can and cannot see. It is a marginal bounding-box test: it verifies that each treated stand's covariate *ranges* fall within the control ranges of its own fireshed, which is a necessary condition for a well-defined contrast, but it does not verify *joint* support and cannot detect holes in the interior of the covariate space, where a treated stand might have no true control analog even though each of its coordinates is individually covered. I therefore call the result marginal within-fireshed support and decline to claim that positivity is a non-issue; the full overlap scatter is in the appendix (Section D, Figure 5).[4]

5.4 The Coefficient Path: Confounding-Removal vs. Overcontrol

The single most informative object for identification is the path the estimate traces as controls enter, and it is worth reading Table 3 a second time with that in view. The pattern is the one Plantinga, Walsh, and Wibbenmeyer read off their own column movements (Plantinga et al., 2022): the coefficients diminish in magnitude as physical controls are added, and the direction of that diminution tells you which way the omitted variables were biasing the raw estimate. Here the raw hazard ratio is already protective at 0.55, and adding the siting confounders and then the fixed effects moves it *toward* the null, to 0.7166, plotted in Figure 3. The estimate travels up, not down, as adjustment proceeds.

[4]The overlap diagnostic is computed over 201 firesheds and the fireshed-clustered inference of Section 5.1 over $G = 296$ fireshed groups; the two are different coarsenings of the same layer, and I carry each with its own count rather than reconcile them silently.

That direction is the crux. When harvest is sited on land that is intrinsically less fire-prone—flatter, more accessible, more merchantable, and for those same reasons less likely to burn—the raw comparison *overstates* the protection, because it credits harvest with a gap that was there in the land before any cut. Conditioning on the structure and terrain that drive both the siting decision and the fire propensity removes part of that gap, and the estimate attenuates toward one. This is the ordinary signature of confounding being removed, and it is the *opposite* of the pattern that would worry a reader most: an effect "manufactured by adjustment," in which the protective sign is absent in the raw data and only *emerges* as controls are piled on, would show the estimate moving away from one as it is adjusted. It moves toward one. The protective sign is a feature of the raw data that adjustment shrinks, not a feature that adjustment conjures.

Attenuation-toward-null under the observed confounders does not, by itself, prove that the residual protection is causal, and I do not read it that way. An unobserved same-signed siting confound—favorable siting not captured by SCANFI structure, terrain, or the fireshed fixed effect—would leave a residual protective bias that no amount of the *observed* controls can reach, and the coefficient path is silent about it by construction. What the path does rule out is the manufacture story; what it does not rule out is a residual unobserved confound, which is why the bounding exercise of Section 5.6 and the pre-trend discussion of Section 5.5 carry the rest of the weight. One reassurance the performed battery does offer: adding a prior-fire indicator, the covariate most likely to act as a collider and pull the estimate the other way, leaves it essentially unchanged at 0.719 (Section C), so the estimate is not balanced on a knife-edge of that one control.

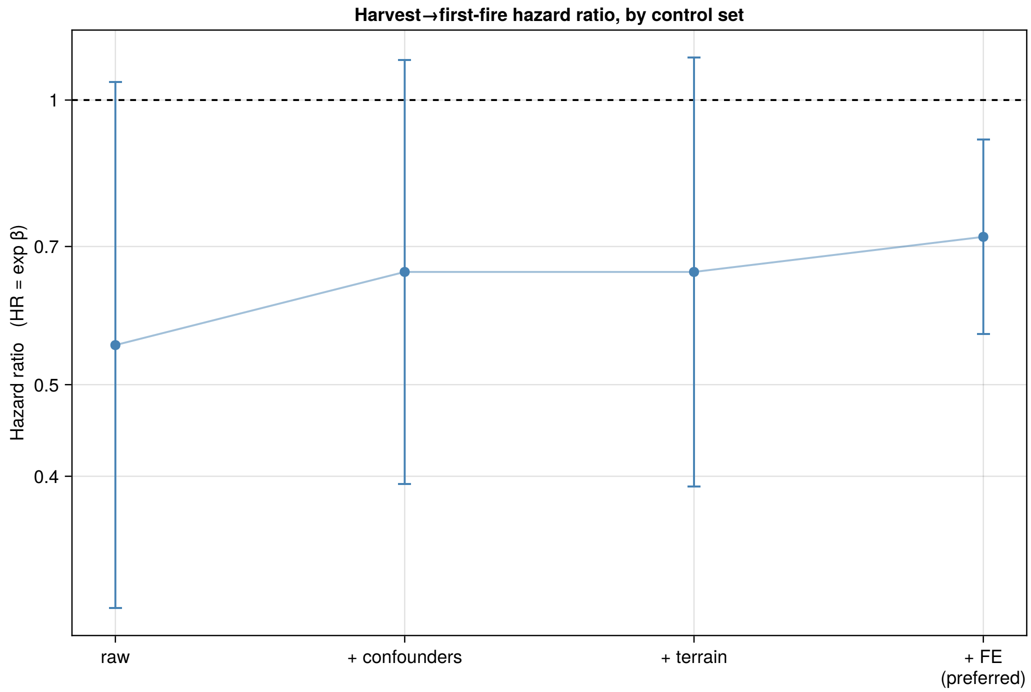

Figure 3: The coefficient path of the pooled harvest–wildfire hazard ratio as controls enter cumulatively: raw / unadjusted, + Tier-A siting confounders, + terrain precision terms, and + ecozone×year and coarsened-fireshed fixed effects (the preferred specification). Points are the hazard ratio $\exp(\beta)$; whiskers are the 95% fire-year cluster-robust confidence interval; the dashed line marks HR $= 1$ (no association). The vertical axis is on a log scale. The estimate moves *up*, toward the null, as siting confounders and fixed effects enter.

5.5 Pre-Trend and Placebo Leads: the Identification Crux

A parallel-trends design lives or dies on the pre-treatment leads, and this is where I am most exposed. The clean, closest lead is the reportable pre-trend statistic. The $\tau = -1$ lead coefficient is 0.204 on the cloglog scale, a hazard ratio of 1.23 with a 95% confidence interval of [0.77, 1.95], which does not reject conditional parallel trends at $\tau = -1$: the year before harvest, stands that will be cut are not detectably diverging from those that will not. On its own terms that is the check one wants, and it comes back clean (Roth et al., 2023).

The leads at longer horizons cannot be read the same way, and I am explicit that they are an artifact rather than a test. The far-lead coefficients run monotonically from a hazard ratio near 0.007 at $\tau = -5$ to near 0.13 at $\tau = -2$ —absurdly, mechanically protective—and they are so for two compounding reasons that have nothing to do with an economic pre-trend. The first is immortal-time survivorship. A stand enters the treated-lead arm only if it is in fact harvested at its eventual date and survives fire-free until then, so the lead window is populated by definitional survivors; and because the leads panel's reference arm is never-treated stands only, carrying no event time of its own, it cannot difference out the resulting survivor gradient. The second is salvage-window contamination. The five-year salvage exclusion that keeps post-fire salvage harvest out of the treatment is coextensive with the lead window, so it mechanically strips pre-harvest fires from the treated arm; and because the salvage filter keys on CanLaD alone, a small share of NBAC pre-harvest fires leaks into the treated arm as well. Together these produce the steep, monotone far-lead gradient, which I disclose as the artifact it is and do not offer as a joint parallel-trends test. The clean $\tau = -1$ lead sits outside the worst of the gradient and is the statistic to read.

What this leaves me is a claim held deliberately at a direction. The protective association survives observable adjustment (Section 4.2), every clustering scheme (Section 5.1), and the removal of the extreme fire-years (Section 5.2), and the clean $\tau = -1$ lead does not reject conditional parallel trends. That is a body of evidence consistent with harvest lowering a stand's subsequent fire hazard—the reading is now available, and I state it once. But because the far-lead immortal-time gradient blocks a genuine *joint* pre-trend test, I cannot certify that the pre-treatment hazards were parallel across the whole lead window, and so a same-signed siting confound that observable adjustment cannot reach, potentially a material share of the estimated protection, is not excluded. The net sign of the residual bias is itself genuinely ambiguous: suppression prioritized near merchantable timber would bias the estimate down, toward over-protection, while the roads and ignition access that harvest brings would bias it up. I neither over-rescue the result nor understate it: it is a robust protective direction, and it is not yet a fully identified effect.

The path from one to the other is a definite program, and I set it out here as a forward agenda rather than apologize for its absence; the build is deferred to Section 7.1. Three steps close the gap, in order. First, a symmetric not-yet-treated leads reference, so that the treated arm's immortal-time conditioning is shared by the control arm and cancels in the contrast, converting the leads into a genuine joint pre-trend test. Second, a salvage-filter fix—taking the union of the CanLaD and NBAC fire records, or reading a clean lead from outside the five-year salvage window, at $\tau \leq -6$—to remove the salvage-adjacency contamination. Third, re-estimation of a clean, powered, out-of-window, immortal-time-cancelled lead: if it sits at zero, the claim moves from direction to identified effect. Until that runs, the ceiling stands where I have set it.

5.6 Sensitivity and Bounding

The natural way to make the residual-confound worry concrete is to ask how strong an unobserved confounder would have to be to overturn the result, and the E-value of VanderWeele and Ding (VanderWeele and Ding, 2017) answers exactly that. On the pooled estimate, an unmeasured confounder would have to be associated with both harvest siting and first-fire hazard by a risk ratio of at least 2.14—over and above everything the observed siting covariates already capture—to explain the point estimate away entirely, and by at least 1.43 to move the confidence bound to the null. I read these as bounding targets rather than as vindication. The E-value is the appropriate sensitivity statistic here because it is valid for a rare-event outcome, which a first fire in a given stand-year is. A threshold of 2.14 is comparable to, not far above, the associations the observed covariates themselves carry with fire risk, which is precisely why a same-signed siting confound that observable adjustment cannot reach—potentially a material share of the estimated protection—

cannot be ruled out (Section 5.5): a confound of that strength is not exotic, and the design has not yet excluded it. The bound and the pre-trend verdict agree, and they set the paper's honest ceiling together.

5.7 Heterogeneity by Ecozone

A single national hazard ratio invites the question of whether it describes the whole country or one corner of it, and the natural way to look is to refit the preferred specification inside each ecozone and read the map. Doing so is more disclosure than test, because the fire record thins quickly once it is cut ten ways: of the ten ecozones, only four carry enough treated-and-burned events to be estimated reliably, and I report those four in Figure 4 and set the rest aside rather than read noise into them. The one ecozone estimated with real precision is the Boreal Shield, the dominant boreal fire regime, at a hazard ratio of 0.545 with a 95% confidence interval of [0.360, 0.826] —below the national pooled value and well clear of one—resting on roughly 15,400 first-fire events, the largest count of any ecozone. That the national number and the Boreal Shield number sit so close together is not a coincidence: the boreal is where most of the country's fire and most of its harvest coincide, so the pooled estimate is anchored by, and largely a reflection of, this one regime.

The other three estimable ecozones lean the way the national result does without resolving on their own. Montane Cordillera and Boreal Plains come in protective at hazard ratios of 0.82 and 0.85, and Taiga Plains nominally adverse at 1.17, but all three intervals include one, so none of them individually adjudicates anything. What matters for reading the national estimate is the shape of that spread rather than any one point in it: no ecozone reverses the sign with precision, and in particular nothing adverse clears its own interval. The protective effect neither holds uniformly across the map with precision—most non-boreal ecozones are simply too fire-sparse at the harvested margin to say—nor reverses cleanly anywhere; it is concentrated in and carried by the boreal, and the ecozone cut is texture on the national result rather than a revision of it.

The six ecozones I exclude are excluded by a rule fixed in advance, so that their omission is not mistaken for the suppression of an inconvenient pattern. Two—Taiga Cordillera and Hudson Plains—are quasi-separation artifacts: within their fixed-effect cells harvested stands essentially never burn inside the window, so the cloglog likelihood runs the log-hazard-ratio to the boundary and the fitted hazard ratio collapses toward zero. I flag any fit with $|\log \mathrm{HR}| > 3$ as separated and read neither the collapsed point nor its interval as an effect. The other four—Taiga Shield, Boreal Cordillera, and the Pacific and Atlantic Maritime zones—have standard errors on the log-hazard-ratio scale above 0.5, wide enough that their intervals span an order of magnitude, so I set them aside as too imprecise to interpret. Their exclusion is, if anything, conservative for the national reading: the two separated zones collapse in the protective direction, and of the four wide-interval zones two sit nominally protective with intervals entirely below one (Taiga Shield at a hazard ratio of 0.08, Boreal Cordillera at 0.05) while only two straddle it. A note on the sparsity: fire-year clustering yields thirty-nine clusters in every ecozone, because the calendar span is full whichever ecozone one conditions on, so the binding constraint on precision is the count of treated-and-burned events and the incidence of fixed-effect separation, never the number of clusters.

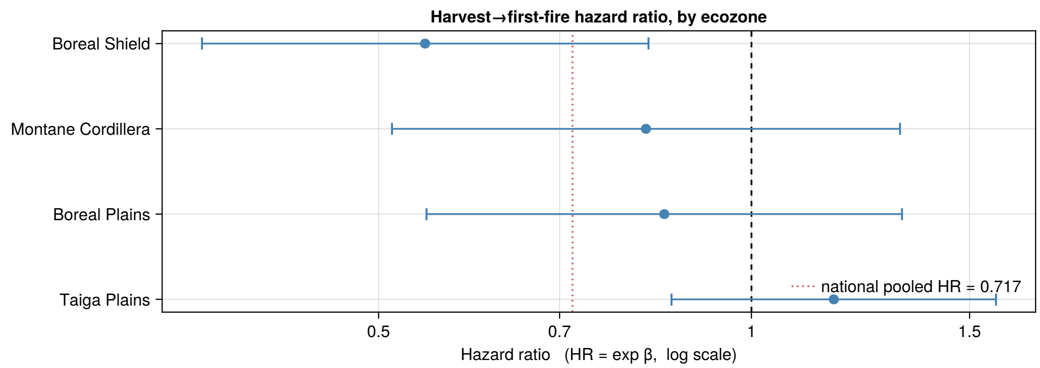

Figure 4: Harvest–wildfire hazard ratio estimated separately within each ecozone, for the four ecozones with a reliably estimable fit. Each point is the ecozone-specific hazard ratio from the preferred complementary-log-log specification refit on that ecozone alone; the horizontal bar is its 95% fire-year cluster-robust confidence interval; the horizontal axis is on a log scale. The dashed vertical line marks HR = 1 and the dotted line the national pooled hazard ratio (0.717). Boreal Shield, the dominant boreal fire regime, is the single precisely estimated ecozone (HR 0.545, interval [0.360, 0.826]) and sits below the national value; the other three credible ecozones lean protective or mildly adverse with intervals that include one. Six further ecozones are omitted as non-interpretable—four with standard errors large enough that their intervals span an order of magnitude, and two with quasi-separation artifacts in which harvested stands almost never burn inside the window within their fixed-effect cells. The omitted zones, if anything, lean protective.

6 Mechanisms

The event-time profile has a shape, and the shape is what carries whatever mechanism reading the data will support. Two channels are usually set against each other in the forest-ecology literature, and they pull in opposite directions. The first is a young-fuel channel: a cut is followed by dense, even-aged regeneration and a flush of fine surface fuel and slash, and that young structure can carry fire more readily than the standing timber it replaced, so harvest would *raise* the hazard—most visibly in the years soon after the cut, when the slash is freshest and the regeneration thickest (Lindenmayer et al., 2009). The second is a reduced-load channel: removing the standing biomass removes the fuel a crown fire needs, so harvest would *lower* the hazard, which is the logic that underwrites deliberate fuel-reduction treatment (Agee and Skinner, 2005; Brodie et al., 2024). Which force dominates over the horizon I observe is an empirical question, and the profile of Section 4.1 is the evidence that bears on it.

What the profile shows is a protective association almost everywhere and no early positive bump. The hazard ratio is below one at 27 of 29 horizons, and the near arms—$\tau = 2$ through $\tau = 8$, the window in which a young-fuel spike would be most visible—are all protective, if individually imprecise. No horizon has harvested stands burning detectably more. Were the young-fuel channel the governing force, its natural signature would be an early interval of elevated hazard that fades as the regeneration matures and self-thins; the data show the reverse ordering, a gentle protection early that *deepens* rather than reverses, with the single most protective horizon out at $\tau = 28$ (HR 0.33). Read at face value, that shape is more consistent with the reduced-load channel dominating over the post-treatment window than with a young-fuel spike—suggestive, in the field's sense of the word, of removed standing load rather than fresh regeneration setting the sign.

I hold that reading at arm's length, for two reasons. The first is the ceiling that governs the whole paper: the profile describes an *association*, and its mechanism interpretation inherits the unresolved siting confound (Section 5.5). A protective shape that partly reflects harvest being sited on land that would in any case have burned less is a shape about siting, not about fuel, and the deepening at long horizons—where the treated sample is thinnest and least powered—is exactly where an unobserved siting gradient would be hardest to tell from a fuel legacy. I do not read the late-τ deepening as a strengthening physical mechanism, and I do not present the profile as having isolated a fuel channel.

The second reason is that a clearcut is not a designed fuel treatment. The reduced-load logic is drawn from silviculture that removes ladder and surface fuel on purpose; clearcutting, which dominates Canadian harvest (Appendix E.1), takes the merchantable stems and leaves slash behind, and its net effect on flammability is not a foregone conclusion in the direction that fuel-treatment theory would predict for a purpose-built treatment. The controlled boreal evidence is itself a caution—crown thinning has been found not to slow a fire's rate of spread or reduce its total fuel consumption (Appendix E.2)—so the reduced-load channel is neither automatic nor uniform across the dimensions of fire behavior. That a protective sign nonetheless appears, for harvest as practiced rather than for an idealized treatment, is the fact the profile establishes; the channel that produces it is read tentatively, and an honest share of the "protection" may be favorable siting the design has not yet swept out.

Where a mechanism reading would be sharpened, or overturned, is across the forest rather than pooled over it—clearcut versus partial harvest, boreal versus Pacific and mixed stands, young versus old structure—and a positive early-τ signal in a young-fuel-prone subgroup would be the cleanest evidence for the first channel that the pooled profile, protective throughout, does not provide. That heterogeneity is not estimated here; I set it out with the rest of the forward program in Section 7.1.

7 Conclusion

Across Canada's managed forest, stands that have been harvested carry a lower subsequent first-fire hazard than comparable unharvested stands. The pooled estimate is a hazard ratio of 0.7166 —a first-fire hazard roughly 28% below the unharvested baseline—protective at 27 of 29 event-time horizons, and it survives observable adjustment, every clustering scheme I try, and the removal of the extreme 2023 fire season (leave-2023 HR 0.6307, if anything stronger). What I estimate is the ATT among fire-experiencing cells—the fixed effects drop the fully-separated zero-fire cells, so the estimand is defined over stands in ecozone-years and firesheds that burn at all—and within that population 99.5% of treated stands sit on marginal within-fireshed support, so I report the headline as a national estimate rather than a local one. What I do *not* claim is a fully identified causal effect: the pre-trend machinery that would test conditional parallel trends at the horizons that matter is non-functional as currently built (Section 5.5). The paper's result is a robust protective *direction*, held at exactly that ceiling and no higher.

The direction is worth having because of its scale and its design. Across the national managed forest, harvested stands are associated with a lower subsequent first-fire hazard, estimated on a fire outcome measured independently of the harvest treatment and with inference clustered on fire-years rather than on correlated pixels. I do not read this as proof that clearcutting mitigates fire; I read it as a national, quasi-experimental estimate whose sign is protective and whose measurement and inference are disciplined. The contribution is that scale and that design—the first continental-scale estimate of the harvest–fire relationship in Canada, built on an independent outcome—not a verdict on forest practice.

Whether that verdict follows is a policy question the paper deliberately does not answer. The reduced-form direction speaks to the most contested premise of the fire–forest co-management case (Appendix E.2)—whether harvest as practiced leaves land more or less likely to burn—but a protective sign, even a fully identified one, does not by itself justify a prescription. The joint gains from managing fire and forest together turn on cost recovery, on treatment efficacy under the extreme weather that drives the worst boreal fires, and on a welfare and management model that weighs revenue, suppression cost, and carbon against one another. None of that is estimated here. I hold the policy implication as conditional—on closing the identification agenda below, and on the structural model that would convert a reduced-form hazard shift into a co-management decision rule—and offer the result as motivation for that program rather than as its conclusion.

The shape of the co-management question the result sharpens is worth stating concretely. Where stands permit, commercial or partial harvest could in principle deliver fuel reduction while generating revenue that defrays its own cost, in place of the two current defaults: the clearcut-and-dense-regeneration pattern whose flammability consequence is exactly what this paper measures, and the pay-to-dispose fuel cut that lowers load but returns nothing. A protective sign for harvest as practiced is a necessary condition for the first option to dominate—if harvested land burned more, the joint gain would be illusory—but it is nowhere near sufficient, because the size of the hazard shift, its persistence, and its interaction with the extreme weather that overwhelms fuel treatment all enter the calculation a decision rule would need. What my estimate supplies is the sign of that leading term, protective and bounded; what it withholds is everything downstream of it, which is why the co-management case remains a question the paper motivates rather than answers.

Four limitations bound what I have shown, and I state them as scope rather than apology. First, residual siting and suppression selection remains the open identification threat: the design conditions on observed structure, terrain, and fireshed, but an unobserved same-signed confound—favorable siting, or suppression concentrated near merchantable timber—would leave a protective bias the observed controls cannot reach, and the net sign of that bias is genuinely ambiguous. Second, the first-class response to the suppression channel, a management-zone split, is limited by the absence of a national management-zone raster; I treat that gap plainly rather than assume it away. Third, treatment and outcome are measured, not observed: CanLaD dates harvest to detection and NBAC records perimeters, and both carry the timing and edge error that the $\tau = 1$ quarantine and the independent-outcome design are built to blunt but not to eliminate. Fourth, the fireshed and ecozone-year fixed effects restrict the estimand to fire-experiencing cells on marginal common support, so the headline is an ATT over that population and a marginal, not joint, overlap test stands behind the "national" label.

None of these is fatal to the direction, and each is addressable. The result that Canada's harvested stands carry a lower subsequent fire hazard is robust enough to reorient the question—from whether the protective sign is real to how much of it is causal—and stable enough to survive the tests a skeptic would run first. That is the contribution I claim: a national, disciplined estimate whose protective sign is robust and bounded, and a definite path from direction to identified effect. The last of these I set out now, as a program rather than a promise.

7.1 Planned Extensions

This note is a placeholder, to be deleted when the analyses it lists are complete and folded into Section 5 and the appendix. The identification agenda comes first and in order (Section 5.5): a symmetric not-yet-treated leads reference, so that the treated arm's immortal-time conditioning is shared by the controls and cancels in the contrast; a salvage-filter fix, taking the union of the CanLaD and NBAC fire records or reading a clean lead from outside the five-year salvage window ($\tau \leq -6$); and re-estimation of a powered, out-of-window, immortal-time-cancelled pre-trend test that would move the claim from direction to identified effect if it sits at zero. Two falsification exercises follow: a suppression-gradient rebuttal (T6), splitting the estimate by proximity to suppression priority to bound the defended-timber channel, and a windthrow placebo (T7), which asks whether a natural disturbance that removes standing load without a harvest decision reproduces the protective sign. The inference and sensitivity suite is then completed—a wild-cluster (Rademacher) bootstrap and a national CR2/Satterthwaite correction against the modest cluster count, and Oster's δ and the Cinelli–Hazlett robustness value alongside the E-value already reported (Section 5.6)—and the reduced-form estimate is corroborated with independent estimators (entropy balancing, CEM/PSM, Sun–Abraham, and a fire-year jackknife). Each item strengthens or stresses the direction reported here; none is required to state it.

References

Agee, James K. and Carl N. Skinner, "Basic principles of forest fuel reduction treatments," *Forest Ecology and Management*, 2005, *211* (1–2), 83–96.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess, "Revisiting event-study designs: Robust and efficient estimation," *Review of Economic Studies*, 2024, *91* (6), 3253–3285.

British Columbia Forest Practices Board, "Forest and Fire Management in BC: Toward Landscape Resilience," Special Report SR61, Forest Practices Board, Victoria, BC 2023. `https://www.bcfpb.ca/wp-content/uploads/2023/06/SR61-Landscape-Fire-Management.pdf`.

Brodie, Emily G., Eric E. Knapp, Wesley R. Brooks, Stacy A. Drury, and Martin W. Ritchie, "Forest thinning and prescribed burning treatments reduce wildfire severity and buffer the impacts of severe fire weather," *Fire Ecology*, 2024, *20* (1), 17.

Byrne, Brendan et al., "Carbon emissions from the 2023 Canadian wildfires," *Nature*, 2024, *633* (8031), 835–839.

Callaway, Brantly and Pedro H. C. Sant'Anna, "Difference-in-differences with multiple time periods," *Journal of Econometrics*, 2021, *225* (2), 200–230.

Canadian Forest Service, "National Burned Area Composite (NBAC)," Canadian Wildland Fire Information System, Natural Resources Canada 2024. `https://cwfis.cfs.nrcan.gc.ca/datamart`.

de Chaisemartin, Clément and Xavier D'Haultfœuille, "Two-way fixed effects estimators with heterogeneous treatment effects," *American Economic Review*, 2020, *110* (9), 2964–2996.

Ellis, Tegan M., David M. J. S. Bowman, Piyush Jain, Mike D. Flannigan, and Grant J. Williamson, "Global increase in wildfire risk due to climate-driven declines in fuel moisture," *Global Change Biology*, 2022, *28* (4), 1544–1559.

Goodman-Bacon, Andrew, "Difference-in-differences with variation in treatment timing," *Journal of Econometrics*, 2021, *225* (2), 254–277.

Guindon, Luc, Philippe Villemaire, Michael Béland et al., "A new approach for spatializing the Canadian National Forest Inventory (SCANFI) using Landsat dense time series," *Canadian Journal of Forest Research*, 2024, *54* (7), 793–815.

Hanes, Chelene C., Xianli Wang, Piyush Jain, Marc-André Parisien, John M. Little, and Mike D. Flannigan, "Fire-regime changes in Canada over the last half century," *Canadian Journal of Forest Research*, 2019, *49* (3), 256–269.

Lindenmayer, David B., Malcolm L. Hunter, Philip J. Burton, and Philip Gibbons, "Effects of logging on fire regimes in moist forests," *Conservation Letters*, 2009, *2* (6), 271–277.

Perbet, Pierre, Luc Guindon, David L. P. Correia, Philippe Villemaire, Omid Reisi Gahrouei, and Rémi St-Amant, "CanLaD: The Canadian Landsat Disturbance dataset (annual, 30 m)," Natural Resources Canada / Open Government 2022. https://open.canada.ca/data/en/dataset/902801fd-4d9d-4df4-9e95-319e429545cc.

Plantinga, Andrew J., Randall Walsh, and Matthew Wibbenmeyer, "Priorities and effectiveness in wildfire management: Evidence from fire spread in the western United States," *Journal of the Association of Environmental and Resource Economists*, 2022, *9* (4), 603–639.

Roth, Jonathan, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe, "What's trending in difference-in-differences? A synthesis of the recent econometrics literature," *Journal of Econometrics*, 2023, *235* (2), 2218–2244.

Sun, Liyang and Sarah Abraham, "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects," *Journal of Econometrics*, 2021, *225* (2), 175–199.

Thompson, Dan K., Dave Schroeder, Sophie L. Wilkinson, Quinn Barber, Greg Baxter, Hilary Cameron, Rex Hsieh, Ginny Marshall, Brett Moore, Razim Refai, Chris Rodell, Tom Schiks, Gregory J. Verkaik, and Jessica Zerb, "Recent crown thinning in a boreal black spruce forest does not reduce spread rate nor total fuel consumption: results from an experimental crown fire in Alberta, Canada," *Fire*, 2020, *3* (3), 28.

Tymstra, Cordy, Brian J. Stocks, Xinli Cai, and Mike D. Flannigan, "Wildfire management in Canada: Review, challenges and opportunities," *Progress in Disaster Science*, 2020, *5*, 100045.

VanderWeele, Tyler J. and Peng Ding, "Sensitivity analysis in observational research: Introducing the E-value," *Annals of Internal Medicine*, 2017, *167* (4), 268–274.

A Data Construction Details

The stand-level panel the event study consumes is built in two steps that Section 2 describes in outline and that we record here in full. The first delineates stands. We form them as connected components over the joint (disturbance-year, disturbance-type) labeling of the CanLaD raster, so that a stand is a contiguous patch of land assigned the same disturbance in the same year—the plausible footprint of a single operation. A connected-components pass on a national grid throws off degenerate shapes, single-pixel specks and pixel-thin filaments that trace raster edges rather than real operations, and a set of pathology guards removes these before they can masquerade as stands. Never-treated land carries no disturbance event to delineate it, so we tile it into pseudo-stands drawn from the empirical block-size distribution of the treated stands; this keeps treated and control units area-commensurable, so the two groups differ in treatment rather than systematically in grain, and it fixes the estimand—through area precision weighting—as the effect on the average stand.

The second step brings the covariate layers into a common frame. Treatment (CanLaD), the independent fire outcome (NBAC), forest structure (SCANFI), terrain (the Canadian Digital Elevation Model), and land cover (VLCE2) are native to different projections, grid origins, and resolutions, so a single point on the ground lands at different pixel coordinates in each. We warp every layer onto the CanLaD 30 m national reference grid with GDAL's `gdalwarp`, resampling continuous fields such as elevation bilinearly and categorical fields such as land cover by nearest neighbor, so that a given pixel index denotes the identical patch of ground in every layer. Covariates are then sliced strictly pre-treatment: for a stand harvested in year c, every element of x_i is read from a layer dated before c, so that no post-harvest information enters the conditioning set. The warped, pre-treatment-sliced, stand-aggregated panel is the object Equation (1) is fit to.

B Estimator Details

The headline is a staggered-adoption discrete-time complementary-log-log first-fire hazard with covariate adjustment and two high-dimensional fixed effects, estimated in R with `fixest::feglm` under the `cloglog` link. The first-fire structure lets us aggregate the stand-year risk set to sufficient statistics—counts of at-risk stand-years and first-fire events within each covariate and fixed-effect cell—rather than carry the full 10^{10}-row pixel panel, which is what makes a national fit tractable; the two fixed effects, ecozone-by-year and coarsened fireshed, are absorbed by alternating projection, and the fit converges. An earlier version of the project used a bespoke sparse aggregation device; it has been retired in favor of this standard estimator, against which the aggregation was checked for numerical equivalence on the shared cells.

The fixed effects buy their spatial control at a cost the estimand pays openly. A fixed-effect group that contains no fire at all is perfectly separated in a hazard model: it carries no within-cell outcome variation and contributes nothing to a first-fire likelihood, so it is dropped. In a country where most stand-years never burn, that is a large share of the panel. Table 6 reports the accounting. The ecozone-by-year absorber separates 127 of its 390 groups and the fireshed absorber 86 of its

296, and together they drop 26.9% of stand-years and 29.2% of treated rows from the estimation sample. This is the basis for the "ATT among fire-experiencing cells" scope statement of Section 3.4, and we document it as scope rather than as a defect: the dropped cells are the ones over which a within-fireshed first-fire contrast is not even defined. Inference throughout is cluster-robust on fire-years with a $t(G-1)$ reference ($G=39$), the conservative small-cluster block for season-wide fire shocks; the fireshed and two-way alternatives of Section 5.1 leave the point estimate unchanged and move only the interval width. The wild-cluster bootstrap and national CR2 correction that would further stress the modest cluster count are part of the forward program (Section 7.1).

Table 6: Fixed-effect separation and dropped observations. Rows report, for each fixed effect (ecozone×year and coarsened fireshed) and their combination, the number of groups, the number of fully-separated (zero-fire) groups, and the resulting share of stand-year rows dropped from the estimation sample, with the share of *treated* rows dropped. Perfectly-separated cells—those with no fire and hence no within-cell outcome variation—are removed by the fixed-effect estimator; the combined drop of roughly 27% of stand-years defines the estimand as the ATT among fire-experiencing cells (Section 3.4).

FE dimension	Groups	Separated	Rows dropped	Treated dropped
ecozone×year	390	127	16.7%	19.3%
fireshed	296	86	15.7%	18.4%
combined (lower bound)	—	—	26.9%	29.2%

C Full Robustness Battery

The body carries the decisive leave-out (dropping 2023) and the leave-out that is read asymmetrically is reported here, alongside the one control perturbation that probes the estimate's dependence on a single covariate. Table 7 reports both. Row R2 drops the entire 2018–2024 high-fire window and moves the estimate to a hazard ratio of 0.554; we read it asymmetrically, because dropping that window removes about a fifth of the fire-years and cuts G from 39 to 32, so a *null* in that thinned sample would have been indeterminate rather than a failure, and the strengthened point is consistent with the association without independently confirming it. Row R15 adds the Tier-B prior-fire indicator—the covariate most likely to act as a collider and pull the estimate the other way—and the hazard ratio is essentially unchanged at 0.719, so the collider-risk control is not load-bearing and the headline does not balance on it. The pending falsification and corroboration rows (suppression-gradient rebuttal, windthrow placebo, independent estimators) are named once in Section 7.1 and are not tabulated here.

Table 7: Appendix leave-out and control-perturbation robustness of the pooled harvest–wildfire estimate. Rows report the preferred specification after dropping the 2018–2024 high-fire window (R2) and after adding the Tier-B prior-fire indicator (R15). Columns report the hazard ratio, its 95% fire-year cluster-robust confidence interval, and the t statistic; a * marks a hazard-ratio interval that excludes 1. R2 removes roughly one-fifth of the fire-years and a large share of treated fires, so its strengthened point is read asymmetrically—indeterminate rather than corroborating (Section 5.2); R15 leaves the estimate essentially unchanged, so the collider-risk prior-fire control is not load-bearing (Section 5.4).

Specification	HR	95% CI (HR)	t
leave-2018–24-out	0.554*	[0.37, 0.83]	-3.00
+ prior_fire (Tier-B)	0.719*	[0.57, 0.91]	-2.83

D Additional Diagnostics

The within-fireshed common-support claim of Section 5.3 rests on the fireshed-by-fireshed overlap in Figure 5. Each point is a fireshed, placed by the number of treated stands it holds against the share of those stands on marginal within-fireshed support, and the mass of the distribution sits at the top: 197 of 201 firesheds contribute, holding 99.5% of treated stands, with the four non-contributing firesheds tiny treated-only cells of at most seven stands each. The figure also makes the diagnostic's limit legible. It is a marginal, bounding-box test—it certifies that each treated stand's covariate *ranges* lie inside its fireshed's control ranges, a necessary condition for a defined contrast, but it cannot see holes in the interior of the joint covariate space—which is why we describe the result as marginal within-fireshed support and stop short of declaring positivity a non-issue. The companion pre-treatment covariate-balance figure is carried in the design section (Figure 1), where it motivates the conditioning strategy.

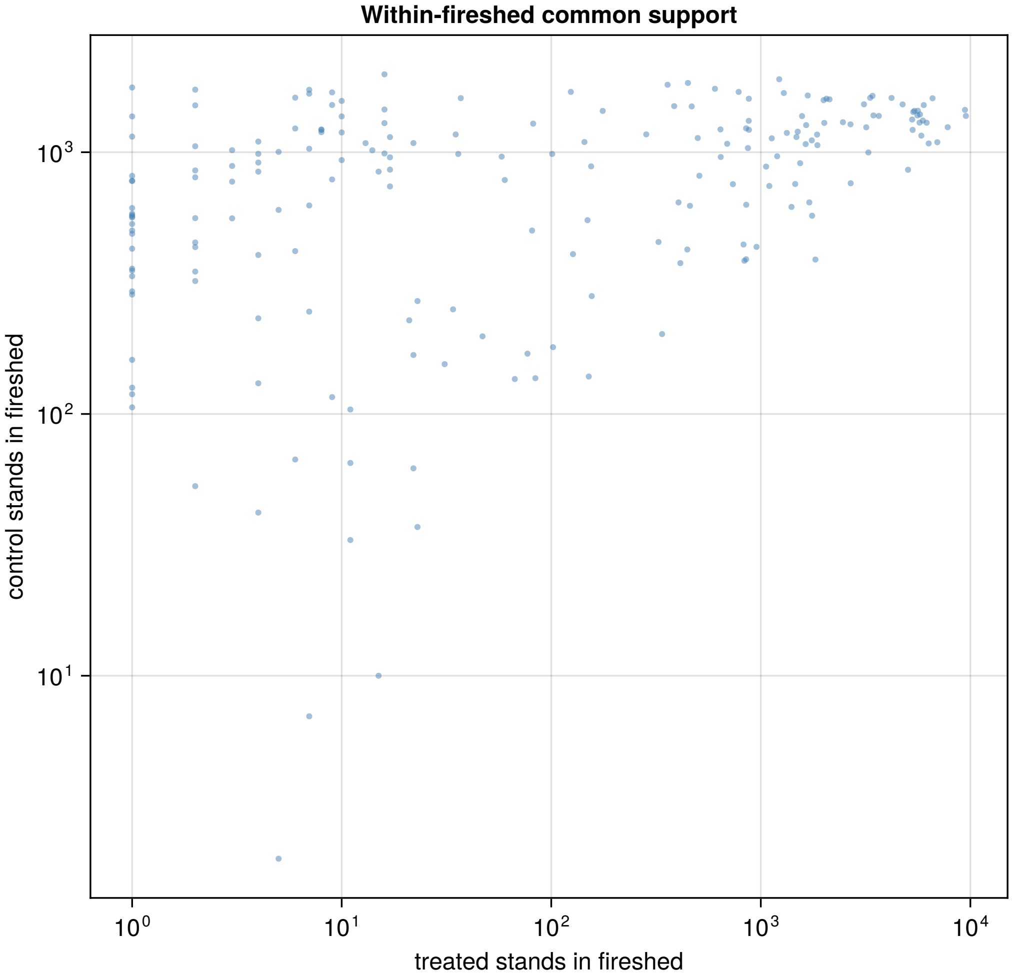

Figure 5: Marginal within-fireshed common support. Each point is a fireshed; the horizontal axis is the number of treated stands and the vertical axis the share of those treated stands whose pre-treatment covariate values fall within the covariate *ranges* (bounding box) spanned by control stands in the same fireshed. 197 of 201 firesheds contribute treated stands on marginal support, covering 99.5% of treated stands; the four non-contributing firesheds hold at most seven treated stands each. This is a marginal test of overlap: it certifies per-covariate range containment, not joint support, and cannot detect joint-support holes.

E Institutional Background: Fire and Forest Management in Canada

E.1 Two Separate Bureaucracies

The policy premise of this paper is a division of labor. Across Canada, wildland fire and timber production are run as operationally distinct programs, with different delivery channels, budgets, and planning horizons, even where they act on the same forest and answer, formally, to the same government. In Ontario, for instance, both functions sit inside the provincial natural-resources ministry, but fire suppression is delivered directly and in-house through a dedicated aviation-and-fire branch, while timber management is delegated to licence holders—forest-products firms, co-operatives, and arm's-length Crown corporations—who prepare and implement the forest-management plans on their units under a Registered Professional Forester's sign-off and Crown approval.[5] Wildland fire is provincial and territorial jurisdiction throughout: there is no federal fire agency, and the national interagency center brokers resource-sharing and mutual aid without holding command over the agencies that actually fight fires (Tymstra et al., 2020). Whatever else it is, the split is durable, and it is the wedge this paper's question exploits.

The split is also old. Fire protection was institutionalized as a distinct administrative branch roughly a century ago, in the provincial forest-fire statutes of the early 1900s, and has stayed operationally separate from timber administration ever since. I read that durability as institutional path-dependence—a boundary drawn when aerial detection and organized suppression were the frontier of fire policy, and left in place as both functions professionalized around it—though I am aware of no source that labels it so, and offer the reading as mine rather than as an established finding. The label matters less than the fact: the boundary is stable, and it did not arise from any judgment that fire and timber are best managed apart.

A second force plausibly reinforces the separation, through the economics of trade rather than the history of administration. The Canada–US softwood-lumber dispute, one of the longest-running trade conflicts between the two countries, turns on the allegation that Canadian provinces subsidize lumber by charging below-market stumpage on Crown land, which supplies the overwhelming majority of the country's fiber; it has produced combined countervailing and antidumping duties on the order of a third of producers' value, and recurrent litigation. This is not a distant sensitivity to fire policy. In a 2026 filing the US industry coalition named Canadian wildfire fuel-management spending and forest-access-road funding among the programs it wants treated as countervailable subsidies. From that we draw only a conjecture, and label it as ours: a government wary of triggering countervailing-duty action has reason to avoid anything that could be cast as subsidizing timber producers, which is exactly what publicly funded fuel-reduction thinning that yields marketable wood would look like. I have found no source that establishes this chilling effect—the documented pattern so far runs the other way, with Canada funding such programs and the US then attacking them—so I offer it as a plausible reinforcing pressure on the separation, not as a demonstrated one.

The two programs run on incompatible clocks, and that is where the separation bites. Timber harvest on public land proceeds through long-horizon forest-management plans—in Ontario, a ten-year planning term layered on tenures that run longer still—prepared years in advance and revised on a decadal cycle. Fire suppression, by contrast, is dispatched operationally and reactively, its horizon a fire season or a single afternoon's weather. Fuel loads and harvest schedules are settled inside the slow planning process; the decision to defend a given stand is taken inside the fast one. Because the two horizons never meet, harvest and fuel are not co-optimized, and there is no institutional

[5] The delegated-licence architecture is Ontario's, not Canada's. Quebec, under its *Sustainable Forest Development Act*, has the province itself prepare the management plans and auctions a fifth to a quarter of public timber through a state marketing board, allocating the balance through guaranteed supply agreements. "Delegated to licence holders" is therefore a provincial fact, not a national one.

venue in which a cut might be timed or shaped to serve a fire objective, or a fire objective allowed to reshape a cut. The separation is not merely administrative tidiness. It is the reason the joint problem is, at present, nobody's problem.

Harvest practice sharpens the point. Clearcutting is the predominant silvicultural system in the Canadian managed forest; partial cuts and commercial thinning are the minority. This matters for what my estimate can mean, because a clearcut is not a fuel-reduction treatment—if anything it is close to the reverse. Deliberate fuel management targets the small-diameter, non-merchantable material that drives fire behavior—ladder fuels, understory, and surface fuels below the roughly 7–12 cm merchantability threshold—in labor-intensive, low-mechanization work that yields little or no timber revenue and is therefore expensive, with published costs spanning from tens to well over a thousand US dollars per acre. The material it removes has, in the typical case, no buyer: it is pile-burned, masticated and left on site, or broadcast-burned, because handling and haul cost exceed the value of dispersed small wood.[6] A clearcut inverts each of these features. It takes the merchantable stems, generates revenue, and leaves behind, over the following years, a dense even-aged regeneration rather than an opened, fuel-poor stand. The sign of the harvest–fire relationship is thus ambiguous on the underlying forest science—young regenerating fuel arguing one way, removed standing biomass the other—and a protective finding, if it holds, cannot be waved through as obvious fuel reduction. It would be an empirical statement about clearcut-dominated harvest as it is actually practiced, not about an idealized treatment that Canadian forestry does not, for the most part, perform.

E.2 Why Wildfire is a Material Economic and Carbon Problem

The stakes that make the question worth answering are, first, atmospheric. The 2023 season, whose 647 million-tonne carbon release opened this paper, amounted to roughly 6% of global fossil-fuel CO_2 that year—set against a global fossil total near 37 billion tonnes—and to more than four times Canada's own fossil emissions (Byrne et al., 2024). A single country's forests, burning, registered at the scale of a major emitting economy. The figure carries two qualifications that I state rather than bury: it is a gross biogenic release that regrowth reabsorbs in part over subsequent decades, so it is not interchangeable with a permanent fossil-fuel tonne, and 2023 was an extreme outlier, on the order of seven times the typical year's area burned. Both bear on interpretation, and neither shrinks the season to insignificance. That singular character is also why the year earns its own robustness check: because 2023 is one cross-fireshed mega-shock rather than many independent events, I ask throughout whether the protective association is merely its shadow, and drop it to find out (Section 5.2).

The season was extreme, but the baseline it departed from is itself rising. Fire-regime intensification across Canada over the last half century (Hanes et al., 2019), driven in part by the climate-linked decline in fuel moisture that raises flammability continent-wide (Ellis et al., 2022), means the rele-

[6] Biomass-energy utilization is a genuine exception, but it is confined to the places where the collection and conversion infrastructure already exists.

vant counterfactual is not a stationary risk but a trend. Harvest, against that backdrop, is at most a marginal lever: no plausible change in cutting practice offsets the climatic forcing, and any harvest signal must be identified against fire-years that move together strongly. That correlation structure is not a nuisance to be conditioned away but a first-order feature of the data, and it dictates how I conduct inference—clustered on fire-years, and stress-tested by removing the years that dominate the record.

It is the conjunction of the two—a rising externality and two programs that do not coordinate—that gives the question its economic edge. The efficiency argument for managing fire and forest together is easy to state: where stands permit, commercial or partial harvest could achieve fuel reduction while generating revenue that defrays the cost of treatment, in place of both the clearcut-plus-dense- regeneration pattern that may raise flammability and the pay-to-dispose fuel cut that raises no revenue at all. Whether those joint gains are large, or even reliably positive, is genuinely contested rather than a free lunch. Cost recovery from thinning is usually incomplete; treatment efficacy is overwhelmed by the extreme weather that drives the worst boreal fires; at least one controlled boreal experiment found that crown thinning did not slow a fire's rate of spread, even as it cut fire intensity to roughly a third of the untreated control (Thompson et al., 2020); and a substantial ecological literature holds that logging can raise, not lower, fire severity (Lindenmayer et al., 2009). I raise the co-management case here as motivation, and hold it there. The reduced-form protective direction this paper recovers speaks directly to the last, most contested premise—whether harvest as practiced leaves land more or less likely to burn—but it does not by itself settle the policy question, which would turn on a welfare and management model I defer to future work (Section 7). What the separation and the stakes together establish is narrower and sufficient: the sign of the harvest–fire relationship is worth knowing, and no institution is currently charged with learning it.